Schooling and college inside the adjusted CH index

Controls: CH specification—County-adjusted CH; Geography + human capital.

1. The county productivity adjustment. Let q_c, n_c and a_c denote county GDP, employment and land area, with ordinary employment density $d_c = n_c/a_c$. Let S_c be average years of schooling and C_c the college share between zero and one. Define $h_c = \beta S_c$, using the fixed micro-based schooling return $\beta = 0.102623$. The college coefficient is fixed at $\kappa = 0.75$. Geography coefficients are reestimated from

$$\log(q_c/n_c) - h_c - \kappa C_c = \alpha_{s(c)} + \rho_C \log d_c + X_c'\gamma + \lambda \log a_c + u_c.$$

The left-hand side removes the specified schooling and college contributions, analogously to the ordinary County IV adjustment. Modern state fixed effects are $\alpha_{s(c)}$; X_c contains geography and transport controls. Log county area is optional and off by default. Under IV, historical population in levels instruments ordinary log employment density.

2. Combine the adjusted county contributions. For MSA or state g, let $N_g = \sum_{c \in g} n_c$. The combined index is

$$D_g^{\text{geo+HC}}(\theta) = \frac{1}{N_g} \sum_{c \in g} \exp(X_c'\widehat{\gamma} + h_c + \kappa C_c) n_c^{\theta} a_c^{1-\theta}.$$

Geography, schooling and college multiply each county's contribution before aggregation. Employment density and the denominator use actual employment. Schooling is not multiplied by θ. The county density estimate ρ_C, state fixed effects and optional area coefficient do not enter this multiplier.

3. The regional outcome. The regional regression uses log GDP per employment and the combined log index, estimating $\rho = \theta - 1$. For all industries the outcome is $\log(Q_g/N_g)$, where Q_g is total GDP; an MSA sector choice uses that sector's GDP and employment. No schooling or college subtraction is made from this outcome. Here each county's human capital enters jointly with its density and geography inside the sum, rather than as a correction calculated from MSA averages.

4. What is estimated and held fixed. Geography coefficients are estimated anew for this specification. The schooling return and college coefficient are supplied constants, not additional coefficients fitted to the regional data. Standard errors account for estimated geography and shared county–regional observations, but treat those two constants as fixed. Calibration uses eligible counties in represented states, including nonmetropolitan counties. Every constituent county enters the index; a missing required input in a positive-employment county excludes the whole region. Consequently, adding human capital may also change the sample, which should be checked when comparing estimates.